Oxford Cambridge and RSA – style practice paper

GCSE (9–1) Computer Science

J277/01 Computer systems

Topic 1.5: Systems software

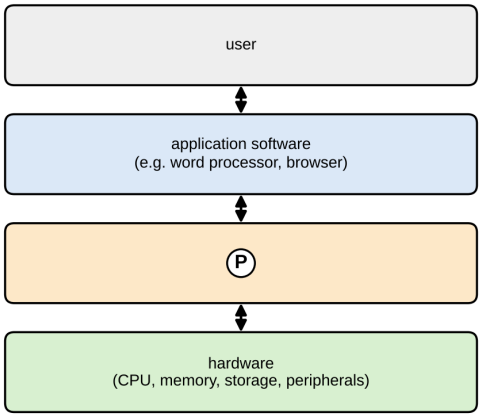
Candidate forename		**Candidate surname**	
Centre number		**Candidate number**	

Practice questions in the style of OCR GCSE Computer Science (J277). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 80 minutes (roughly 1 minute per mark)
Instructions	• Use black ink. HB pencil may be used for graphs and diagrams only. • Answer **all** the questions. • Write your answer to each question in the space provided. • Where the question asks you to write code or pseudocode, you may use any suitable high-level language or the OCR Reference Language; marks are for the logic, not exact syntax. • Do **not** use a calculator or any other electronic device unless told to.
Information	• The total mark for this paper is **72**. • The marks for each question are shown in brackets [ ]. • Quality of extended responses will be assessed in questions marked with an asterisk (*). • Where a question asks you to *state* or *give*, a short answer is enough; *describe* and *explain* need more detail.

Name: _____ Date: _____ Mark: _____ / 72

1 The diagram shows how a computer system is organised. Layer **P** is the operating system (OS).



(a) State what is meant by an **operating system**. **[2]**

(b) Give **one** example of an operating system. **[1]**

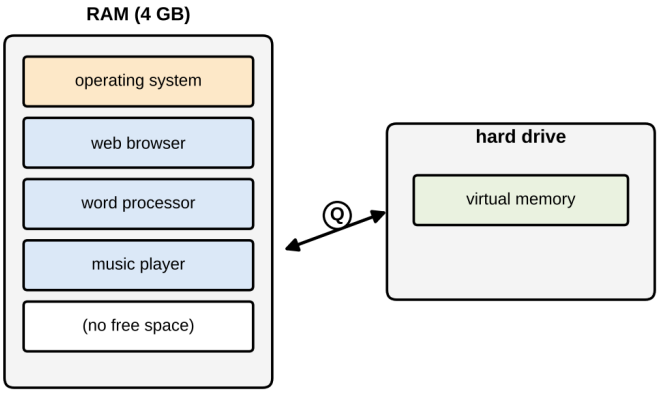
(c) Draw **one** line from each function of the operating system to its description. **[4]**

Function	Description
User interface	Uses drivers to communicate with printers and keyboards
Memory management	Creates, names, moves and deletes files and folders
Peripheral management	Lets the user interact with the computer using icons or commands
File management	Allocates RAM to each running program

(d) Describe the purpose of a **device driver**. **[2]**

(e) A user plugs in a new printer but it does not print. Suggest why and what the user should do. **[2]**

2 Computers can run several programs at the same time.



(a) State the name given to running several programs at once. **[1]**

(b) Explain how the operating system allows multitasking. **[3]**

(c) The diagram shows the RAM of a computer that is full. The user opens another program. Name the process **Q** that allows the new program to run. **[1]**

(d) Describe what happens in **Q**. **[2]**

(e) Explain why having a large amount of RAM improves the performance of a computer that runs many programs. **[2]**

3 A school's operating system manages users and files.

(a) Describe **two** functions of user management in an operating system. **[2]**

(b) Explain why students and teachers are given different access rights to the school's files. **[2]**

(c) Describe how an operating system helps a user to organise their files. **[2]**

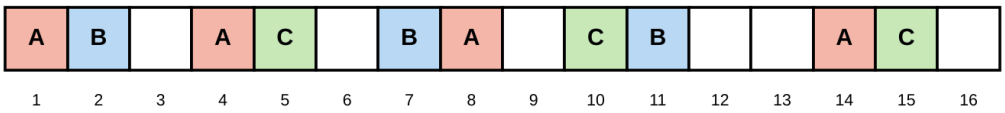
(d) State what a **graphical user interface (GUI)** is and give **one** advantage of using a GUI instead of a command line interface. **[2]**

4 Utility software helps to maintain a computer.

(a) State what is meant by **utility software**. **[1]**

(b) Name **three** types of utility software. **[3]**

(c) The diagram shows part of a hard disk drive. Each square is a block of storage. Letters show where parts of the files **A**, **B** and **C** are stored; empty squares are unused.



(d) Into how many separate pieces (fragments) is file **A** split? **[1]**

(e) Explain why this makes the disk slower to read. **[2]**

(f) Describe how **defragmentation** software improves performance. **[3]**

(g) Explain why defragmentation is not needed on a solid state drive. **[2]**

5 A company uses utility software to protect its data.

(a) Describe what **anti-malware** software does. **[3]**

(b) State what is meant by **encryption software** and why it is used. **[2]**

(c) Explain how **backup software** helps a company if its main storage fails. **[2]**

(d) Describe the difference between a **full backup** and an **incremental backup**. **[2]**

(e) Explain why a company might compress its files before backing them up. **[2]**

6 Software can be classified as system software or application software.

(a) Tick (✓) **two** boxes to show which are examples of **application software**. **[2]**

- ☐ Operating system
- ☐ Spreadsheet
- ☐ Defragmentation tool
- ☐ Web browser
- ☐ Device driver

(b) Give **one** difference between system software and application software. **[1]**

(c) Explain why a computer needs an operating system before application software can be used. **[2]**

7 Extended response questions.

(a) *Discuss the role of the operating system in managing the hardware and software of a computer. In your answer you should refer to memory management, peripheral management and user management. **[8]**

(b) *A small business has an old computer that is running slowly and has had several virus infections. Discuss the utility software and other steps the business could use to improve performance and security. **[8]**

END OF QUESTIONS

MARK SCHEME – Topic 1.5: Systems software

How to use this mark scheme: mark your own work in a different colour. A bullet (•) in the Answer column is a separate marking point worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed. **Accept** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. For programming questions, any valid solution that works is acceptable and syntax errors are not penalised if the intent is clear. The code in the answers uses the OCR Reference Language style.

Question	Answer	Marks	Guidance
1(a)	• system software that manages the computer's hardware and software resources ; • provides an interface / platform for the user and application software to run ;	2 AO1 1.5.1	
1(b)	Windows / macOS / Linux / Android / iOS ;	1 AO1 1.5.1	
1(c)	• user interface: allows the user to interact with the computer ; • memory management: allocates RAM to programs and stops them overwriting each other ; • peripheral management: controls input / output devices using drivers ; • file management: organises files into folders / directories and controls access ;	4 AO1 1.5.1	One mark for each correct match.
1(d)	• software that allows the operating system to communicate with / control a hardware device ; • translates OS instructions into commands the device understands ;	2 AO1 1.5.1	
1(e)	• the correct driver is not installed ; • install / download the driver for that printer / OS ;	2 AO3 1.5.1	
2(a)	multitasking ;	1 AO1 1.5.1	
2(b)	• each program is loaded into / given its own area of RAM ; • the CPU gives each program a short slice of processing time in turn ; • this happens so quickly that the programs appear to run at the same time ;	3 AO2 1.5.1	
2(c)	virtual memory / paging / swapping ;	1 AO2 1.5.1	
2(d)	• data / pages not currently in use are moved from RAM to the hard drive ; • this frees RAM for the new program; data is moved back when needed ;	2 AO2 1.5.1	
2(e)	• more programs / data can be held in RAM at once ; • so less need for slower virtual memory / swapping ;	2 AO3 1.5.1	
3(a)	• creates / deletes user accounts ; • controls login using usernames / passwords ; • sets access rights / permissions for each user ; • provides each user with a separate profile / settings ;	2 AO1 1.5.1	Any two.
3(b)	• to prevent students viewing / changing confidential data (e.g. exam papers / marks) ; • teachers need more access to do their work ; protects data integrity ;	2 AO2 1.5.1	

Question	Answer	Marks	Guidance
3(c)	• files are stored in folders / directories ; • the user can create, name, move, copy, delete and search for files ;	2 AO1 1.5.1	
3(d)	• a GUI uses windows, icons, menus and a pointer ; • advantage: easier for beginners / no need to remember commands ;	2 AO1 1.5.1	
4(a)	system software that performs maintenance / housekeeping tasks (helps manage, protect and optimise the computer) ;	1 AO1 1.5.2	
4(b)	• anti-malware / antivirus ; • defragmentation ; • backup ; encryption ; compression ; disk checker ; firewall ;	3 AO1 1.5.2	Any three.
4(d)	4 ;	1 AO2 1.5.2	
4(e)	• the read head / drive must move to several places to read the file ; • which takes extra time compared with reading the file stored together ;	2 AO2 1.5.2	
4(f)	• rearranges the blocks of each file so that they are next to each other (contiguous) ; • free space is combined into one area ; • so files are read faster / less movement of the read head ;	3 AO1 1.5.2	
4(g)	• SSDs have no moving parts / read head ; • access time is the same wherever data is stored ;	2 AO3 1.5.2	
5(a)	• scans files / programs for known malicious code (signatures) or suspicious behaviour ; • removes or quarantines the malware ; • runs in the background / updated regularly with new malware definitions ;	3 AO1 1.5.2	
5(b)	• converts data into an unreadable form (ciphertext) using a key ; • so that unauthorised people cannot understand it if stolen ;	2 AO1 1.5.2	
5(c)	• copies of the data are stored (e.g. on another disk / in the cloud) ; • so files can be restored / recovered after failure, deletion or ransomware ;	2 AO2 1.5.2	
5(d)	• full backup copies all files ; • incremental copies only files that have changed since the last backup ;	2 AO1 1.5.2	
5(e)	• compressed files are smaller / use less storage space ; • so backups are quicker to make / need less space or bandwidth ;	2 AO2 1.5.2	
6(a)	• Spreadsheet • Web browser	2 AO1 1.5.1	
6(b)	system software manages / runs the computer; application software carries out tasks for the user ;	1 AO1 1.5.1	
6(c)	• the OS manages the hardware / provides the platform that the application runs on ; • application software uses the OS to access memory, files, input and output devices ;	2 AO2 1.5.1	

Question	Answer	Marks	Guidance
7(a)	**Level 3 (high), 6–8 marks**: Thorough knowledge and understanding of the topic with detailed, accurate and relevant points. A balanced, well-structured answer reaching a justified conclusion where asked. Technical terms used correctly throughout. **Level 2 (mid), 3–5 marks**: Reasonable knowledge and understanding. Some relevant points made, but the answer lacks detail, balance or a clear structure in places. **Level 1 (low), 1–2 marks**: Basic knowledge. A few simple, mostly isolated points with little or no development or linking. **Level 0, 0 marks**: No response or no response worthy of credit. **Indicative content** (credit any other valid points): • OS acts as the interface between the user / applications and the hardware • memory management: allocates RAM to each running program; stops programs interfering with each other; uses virtual memory when RAM is full; allows multitasking • peripheral management: sends / receives data between the CPU and input / output devices; uses device drivers; handles interrupts and buffers • user management: creates accounts, logins, passwords; sets access rights and file permissions; separate user profiles • file management: folders, naming, moving, copying, security of files • provides a user interface (GUI / CLI) and manages multitasking by sharing CPU time • conclusion: without the OS applications could not run / user could not use the hardware	8 AO1 / AO2 1.5.1	
7(b)	**Level 3 (high), 6–8 marks**: Thorough knowledge and understanding of the topic with detailed, accurate and relevant points. A balanced, well-structured answer reaching a justified conclusion where asked. Technical terms used correctly throughout. **Level 2 (mid), 3–5 marks**: Reasonable knowledge and understanding. Some relevant points made, but the answer lacks detail, balance or a clear structure in places. **Level 1 (low), 1–2 marks**: Basic knowledge. A few simple, mostly isolated points with little or no development or linking. **Level 0, 0 marks**: No response or no response worthy of credit. **Indicative content** (credit any other valid points): • anti-malware to scan for and remove viruses; keep it updated; run regular scans • firewall to block unauthorised network access • defragmentation (hard drive) to put file blocks together, faster reads; not needed for SSD • disk clean-up / compression to free storage; uninstall unused programs • backup software to restore data after infection / failure; encryption to protect sensitive data • other steps: install more RAM or an SSD, update OS patches, user training, access rights • judgement: security software prevents data loss; hardware upgrade gives the biggest speed improvement	8 AO1 / AO2 / AO3 1.5.2	

Where marks are lost in Topic 1.5

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **OS functions:** learn the four — user interface, memory management, peripheral management (drivers), file management — plus user management.
- ☐ **Multitasking:** RAM is shared between programs and the CPU switches rapidly between them. Mention **time slices**.
- ☐ **Virtual memory:** uses part of the *hard drive* as extra RAM — it is slower, so it should be a last resort.
- ☐ **Defragmentation:** moves file blocks together. It helps **HDDs** (moving read head) but not SSDs.
- ☐ **Utility vs application:** utility software maintains the computer; application software does a task for the user.
- ☐ **Anti-malware vs firewall:** anti-malware scans *files*; a firewall checks *network traffic*.
- ☐ **Backup:** say where the copy is kept (other disk / cloud) and what it is used for (restoring).
- ☐ **Discuss answers:** cover each function in turn and finish with a judgement.